$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

Let $A = \dfrac{\pi}{3} - \theta, \quad B = \dfrac{\pi}{3} + \theta$

$$\begin{aligned} 4\sin\theta\sin\big(\frac{\pi}{3}-\theta\big)\sin\big(\frac{\pi}{3}+\theta\big) &= 2\sin\theta\left(\cos(-2\theta) - \cos\frac{2\pi}{3}\right) \\ &= 2\sin\theta\left(\cos 2\theta + \frac{1}{2}\right) \\ &= 2\sin\theta\cos 2\theta + \sin\theta \\ &= \sin(3\theta) + \sin(-\theta) + \sin\theta \\ &= \sin 3\theta \end{aligned}$$

(i)

$$\ln 4 + \ln\sin\theta + \ln\sin\big(\frac{\pi}{3}-\theta\big) + \ln\sin\big(\frac{\pi}{3}+\theta\big) = \ln\sin 3\theta$$

$$\implies \cot\theta - \cot\big(\frac{\pi}{3}-\theta\big) + \cot\big(\frac{\pi}{3}+\theta\big) = 3\cot 3\theta$$

When $\theta = \dfrac{\pi}{9}$

$$\cot\frac{\pi}{9} - \cot\frac{2\pi}{9} + \cot\frac{4\pi}{9} = 3\cot\frac{\pi}{3} = \sqrt{3}$$

(ii)

$$4\sin\big(\frac{\pi}{6}-\phi\big)\sin\big(\frac{\pi}{6}+\phi\big)\cos\phi = \cos 3\phi$$

$$4\cos\phi\cos\big(\frac{\pi}{3}+\phi\big)\cos\big(\frac{\pi}{3}-\phi\big) = \cos 3\phi$$

$$4\cos\theta\cos\big(\frac{\pi}{3}-\theta\big)\cos\big(\frac{\pi}{3}+\theta\big) = \cos 3\theta$$

Change θ to ϕ and divide by $(*)$:

$$\cot\theta\cot\big(\frac{\pi}{3}-\theta\big)\cot\big(\frac{\pi}{3}+\theta\big) = \cot 3\theta$$

$$4\sin\frac{\pi}{9}\sin\frac{2\pi}{9}\sin\frac{4\pi}{9} = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\begin{aligned}\operatorname{cosec}\frac{\pi}{9}-\operatorname{cosec}\frac{5\pi}{9}+\operatorname{cosec}\frac{7\pi}{9} &= \frac{1}{\sin\frac{\pi}{9}}-\frac{1}{\sin\frac{4\pi}{9}}+\frac{1}{\sin\frac{2\pi}{9}}\\ &= \frac{\sin\frac{2\pi}{9}\sin\frac{4\pi}{9}-\sin\frac{\pi}{9}\sin\frac{2\pi}{9}+\sin\frac{\pi}{9}\sin\frac{4\pi}{9}}{\frac{\sqrt{3}}{8}}\end{aligned}$$

Let $N=\sin\dfrac{2\pi}{9}\sin\dfrac{4\pi}{9}-\sin\dfrac{\pi}{9}\sin\dfrac{2\pi}{9}+\sin\dfrac{\pi}{9}\sin\dfrac{4\pi}{9}$

$$\begin{aligned}N &= \frac{1}{2}\left(\cos\frac{2\pi}{9}+\frac{1}{2}\right)-\frac{1}{2}\left(\cos\frac{\pi}{9}-\frac{1}{2}\right)+\frac{1}{2}\left(\frac{1}{2}-\cos\frac{5\pi}{9}\right)\\ &= \frac{1}{2}\left(\cos\frac{2\pi}{9}+\cos\frac{4\pi}{9}-\cos\frac{\pi}{9}\right)+\frac{3}{4}\\ &= \frac{1}{2}\left(2\cos\frac{\pi}{3}\cos\frac{\pi}{9}-\cos\frac{\pi}{9}\right)+\frac{3}{4}\\ &= \frac{1}{2}\left(\cos\frac{\pi}{9}-\cos\frac{\pi}{9}\right)+\frac{3}{4}\\ &= \frac{3}{4}\end{aligned}$$

$$\operatorname{cosec}\frac{\pi}{9}-\operatorname{cosec}\frac{5\pi}{9}+\operatorname{cosec}\frac{7\pi}{9}=\frac{N}{\frac{\sqrt{3}}{8}}=\frac{3/4}{\sqrt{3}/8}=2\sqrt{3}$$

This solution was submitted by **Jon**

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